

OMER AKBEN

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PROFESSIONAL SUMMARY

AI Full-Stack Engineer with 6+ years of SDET/QA automation experience. Specializes in building production-grade Retrieval-Augmented Generation (RAG) and agentic workflows using LangGraph, LangChain, and modern LLM APIs (OpenAI, Claude, Vertex AI, Azure AI). Combines strong Python (FastAPI, Django) and TypeScript/React (Next.js, Vercel AI SDK) skills with deep testing and CI/CD expertise to ship reliable, low-hallucination AI applications. Known for system-level thinking: observability, automated evaluations (RAGAS, DeepEval), and shift-left quality baked into full-stack delivery.

CORE SKILLS

Languages: Python, TypeScript, JavaScript, C#, Java, SQL, Bash

AI & LLM: LangChain, LangGraph, OpenAI API, Anthropic Claude, Vertex AI, Azure AI / Microsoft Foundry, Amazon Bedrock, RAG, RAGAS, LangSmith, Arize Phoenix, vector databases (pgvector, FAISS, Pinecone, Weaviate), multi-agent systems (CrewAI, Mastra), prompt engineering, golden datasets

Backend & APIs: FastAPI, Django, Node.js/Express, REST, GraphQL, Pydantic, OpenAPI/Swagger, OAuth2/JWT, ORMs, API versioning

Frontend: React, Next.js (App Router), Vercel AI SDK, streaming UIs (SSE), Tailwind CSS, SSR/ISR, accessibility (a11y), performance optimization (Lighthouse)

Data & Infra: PostgreSQL, SQL Server, MongoDB, Redis (semantic caching, rate limiting), Elasticsearch, schema design, migrations, query tuning

Cloud, DevOps & LLMops: AWS (Lambda, ECS, S3, API Gateway), Azure (Entra ID, DevOps), Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, observability, cost/performance optimization

QA & Test Engineering: Playwright (E2E/API), Selenium WebDriver, Cypress, REST Assured, Postman/Newman, Cucumber/SpecFlow (BDD), JUnit/TestNG/xUnit/MSTest, JMeter, Appium, test strategy, TDD, BDD, shift-left practices

PROFESSIONAL EXPERIENCE

Test Automation Architect (SDET) — Oteemo | Remote | Feb 2025 – Sep 2025

- Led quality engineering strategy for a Fortune 500 client, transforming a greenfield Azure DevOps environment into a unified automation ecosystem and accelerating release velocity across distributed teams.
- Designed containerized (Docker) Playwright and Selenium test infrastructure with environment parity between local and CI, cutting new project setup time from days to hours.
- Built deterministic test patterns (intelligent retries, self-healing locators) and parallelized Jenkins/Azure DevOps pipelines, achieving 98% pass rates on regression suites and reducing pull request feedback time by 40%.
- Mentored 5+ engineers on TDD/BDD and shift-left practices, driving a measurable reduction in production defects through stricter code review standards.

Freelance AI Full-Stack Engineer (Part-time, concurrent with full-time roles) | Remote (Raleigh, NC) | Jan 2024 – Present

Clients: Elon University, Minor Use Foundation, North Glass LLC

- Architected a multi-tenant conversational AI platform for higher education using Next.js (Vercel AI SDK) and FastAPI with streaming responses and Microsoft Entra ID RBAC, supporting 1,000+ users and reducing support response time by 60%.
- Built a production RAG pipeline with hybrid search (pgvector + keyword search) backed by Upstash Redis and semantic caching; validated using golden datasets and RAGAS, improving retrieval precision by 25% while keeping P95 latency under 2 seconds.
- Orchestrated multi-agent workflows in LangGraph for autonomous tool calling and student services flows, adding content guardrails and conversation memory to keep interactions compliant with university data policies.
- Automated regulatory document workflows for the Minor Use Foundation and other clients using LLM-based extraction on unstructured PDFs, reducing manual data entry by 60% and implementing audit trails; integrated Playwright E2E tests into GitHub Actions and LangSmith tracing to achieve 95%+ stability on critical chat/auth flows and cut query costs by 15%.

QA Engineer — Engineering Consulting Services (ECS) | Raleigh, NC (Hybrid) | Apr 2023 – Jan 2024

- Refactored core test utilities and helpers, introducing Builder/Strategy patterns for test data and behaviors, increasing team velocity by 20% and reducing code duplication by 35%.
- Converted manual API validation into automated Postman collections with JavaScript assertions and Newman runs in Azure DevOps CI, reducing manual testing backlog by 10% and improving regression repeatability.
- Modernized legacy Selenium suites by migrating to Page Factory with resilient locators and custom waits, boosting UI automation stability from 65% to 95% and cutting maintenance effort by 30%.

SDET / QA Test Automation Engineer — Xsolis | Remote | Oct 2022 – Apr 2023

- Automated healthcare workflows including HL7 message processing, PHI ETL pipelines, and clinical documentation validation using Selenium WebDriver and SpecFlow (BDD with Gherkin), increasing automated coverage from 40% to 85% on critical compliance workflows (PCI, SOX, patient admission, authorization).
- Implemented extensive SQL-based database validation across 50+ tables to ensure data integrity and business rule enforcement, and created automated reconciliation reports comparing source systems with ETL outputs.
- Collaborated with external vendor (EPAM) to align frameworks and coding standards, contributing to shared xUnit suites and Postman collections that accelerated cross-team regression and API contract testing.

SDET / QA Automation Engineer — Fannie Mae | Remote | Mar 2021 – Sep 2022

- Built a Java-based automation framework with Selenium WebDriver, TestNG, and REST Assured covering UI, API, and DB validations for mortgage processing modules (loan origination, underwriting, servicing), reaching 20% automated coverage and improving release velocity by 10%.
- Architected CI test infrastructure with Selenium Grid supporting 50+ concurrent browser sessions across OS/browser combinations; parallelized execution and dynamic test distribution reduced regression time by 15%.
- Implemented an API contract testing suite in REST Assured integrated into CI with automatic failure notifications, and engineered AWS EC2/ECS-based deployment strategies for scalable, containerized test execution.

QA Test Engineer — Delaware State Chamber of Commerce | Wilmington, DE (Hybrid) | Sep 2019 – Mar 2021

- Performed functional, regression, and UAT testing on web applications across multiple sprints, documenting and managing defects in Jira and collaborating with development and product teams to prioritize fixes.
- Developed a Java-based Selenium/JUnit framework integrated with Jenkins CI/CD and optimized SQL validation scripts, increasing automated coverage by 35%, reducing API defects by 30%, and cutting execution time by 10%.

PROJECTS

AI Studio Platform (Elon University) — Next.js/TypeScript • FastAPI • OpenAI/Claude • LangChain/LangGraph • RAG (pgvector/FAISS) • Microsoft Entra ID • Playwright

- Multi-tenant conversational AI platform with agentic workflows (tool/function calling, conversation memory, content guardrails) and RBAC on Entra ID.
- Integrated RAG evaluation (RAGAS) and Playwright E2E/API tests into CI, tracking answer accuracy, context precision/recall, latency percentiles, and visual regressions.

Portfolio (omerakben.com) — Next.js/TypeScript • Vercel • Vercel AI SDK • Mastra • Upstash Redis • Playwright

- AI-native portfolio app using Next.js 15 and React 19 with Mastra agents and Vercel AI SDK; Upstash Redis for rate limiting and vector storage.
- Established quality gates (Vitest, Playwright, Axe-Core), performance budgets (size-limit), and Lighthouse CI + Sentry + PostHog monitoring, sustaining 95+ scores for Performance, Accessibility, and SEO.

tuel (npm packages) — React • TypeScript • Storybook • CI/CD

- Open-source, tree-shakable React animation component library with type-safe APIs, Storybook docs, CI-based testing, and automated npm publishing; used in production on personal and demo sites.

EDUCATION & CERTIFICATIONS

Nashville Software School — Full-Stack Web Developer Bootcamp | 2024–2025

Modern web stack (React/Next.js, Python/FastAPI/Django, TypeScript, TDD, CI/CD, Agentic AI Tools for Developers, Cloud Deployment Fundamentals).

TechCenture Academy — Software Development Engineer in Test (SDET) | 2018–2019

Test automation training covering Java/C#, Selenium WebDriver, Cucumber/SpecFlow (BDD), TDD/BDD, and CI/CD integration with Jenkins and Azure DevOps.

Istanbul Okan University — Master of Science, Healthcare Management | 2014–2016

Focus on healthcare information systems, HIPAA/regulatory compliance, and quality improvement.